



Introduction to Machine Learning

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What Is Machine Learning?

- Machine Learning is the science (and art) of programming computers so they can **learn from data**.
- Here is a slightly more general definition:
 - Machine Learning is the field of study that gives computers the **ability to learn without being explicitly programmed**.

A high-level introduction

**Computer
Science**



**Data
Science**



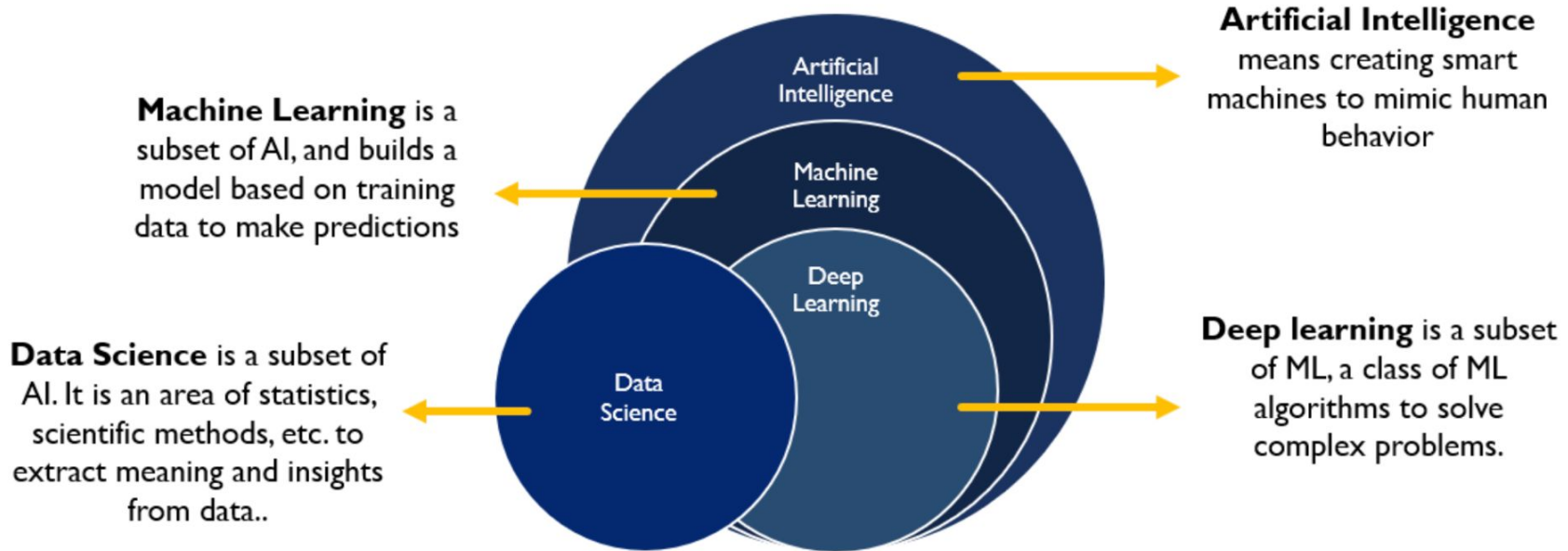
AI



**Machine
Learning**



A high-level introduction



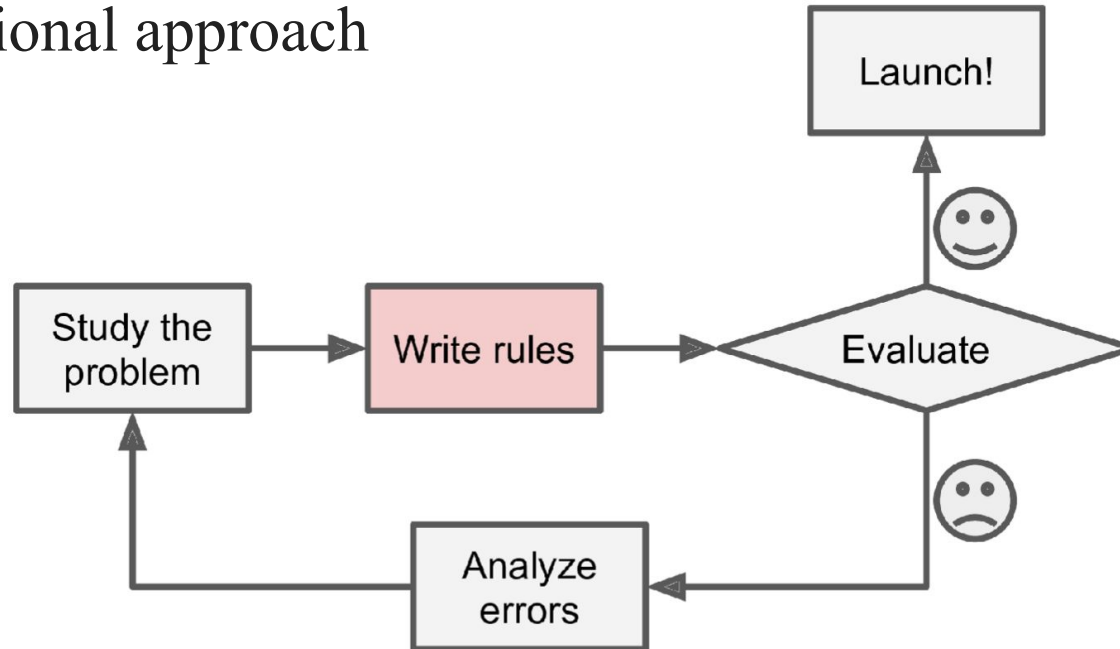


Why Use Machine Learning?

- Consider how you would write a spam filter using traditional programming techniques
 - a. First you would consider what spam typically looks like.
 - b. You would write a detection algorithm for each of the patterns that you noticed.
 - c. You would test your program and repeat steps (a.) and (b.) until it was good enough to launch.

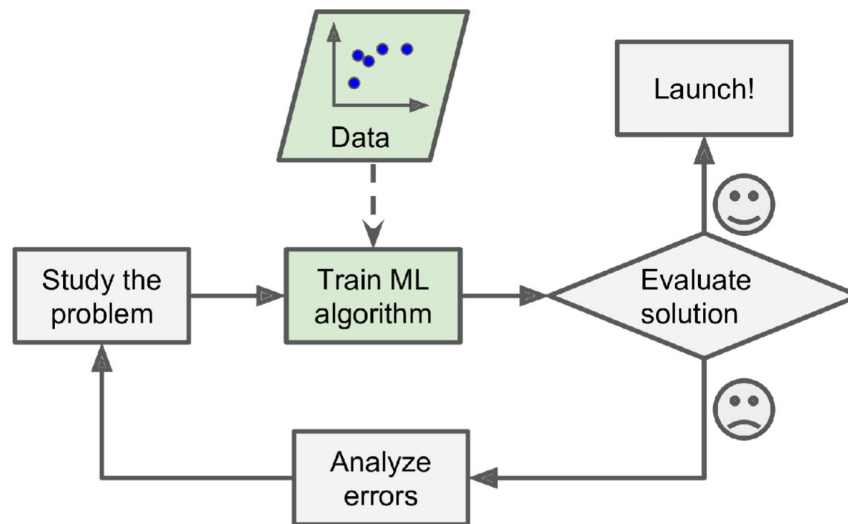
Why Use Machine Learning?

- The traditional approach



Why Use Machine Learning?

- A spam filter based on Machine Learning techniques **automatically learns** which words and phrases.





Machine Learning is great for

- Problems for which existing solutions require a lot of fine-tuning or long lists of rules.
- Complex problems for which using a traditional approach yields no good solution.
- Fluctuating environments.
- Getting insights about complex problems and large amounts of data.



Types of ML Systems

- Whether or not they are trained with human supervision.
 - Supervised, Unsupervised, Semi-supervised, and Reinforcement Learning
- Whether or not they can learn incrementally on the fly.
 - Online versus Batch Learning
- Whether they work by simply comparing new data points to known data points, or instead by detecting patterns.
 - Instance-based versus Model-based Learning

Types of ML Systems

Supervised Learning

- > Labeled data
- > Direct feedback
- > Predict outcome/future

Unsupervised Learning

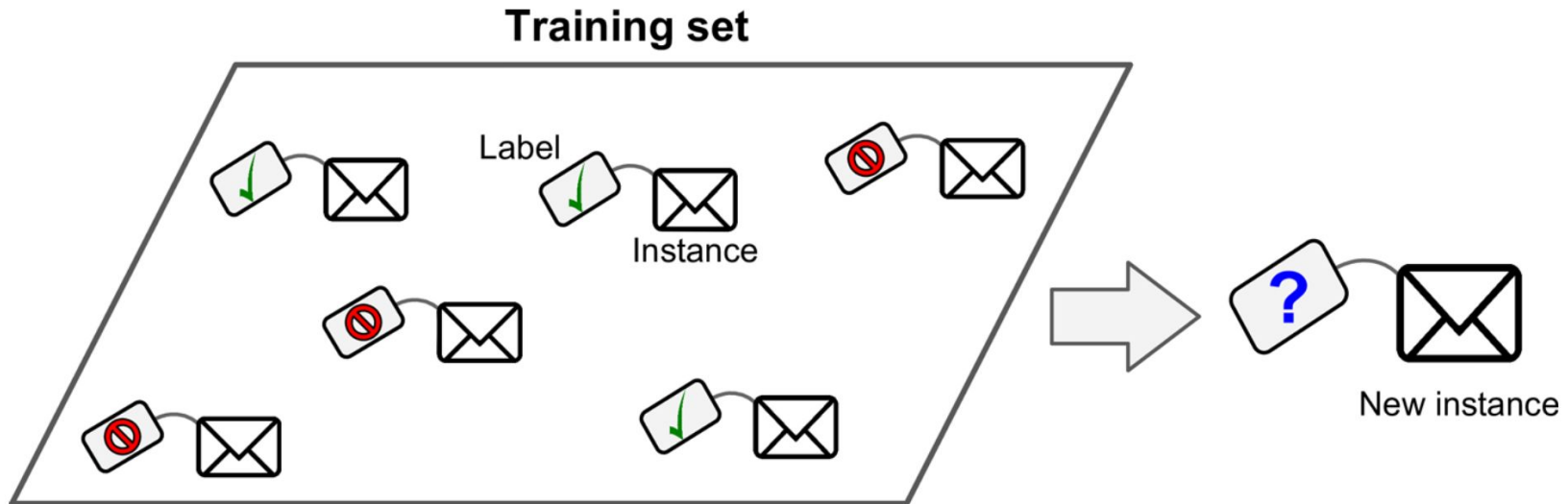
- > No labels
- > No feedback
- > Find hidden structure in data

Reinforcement Learning

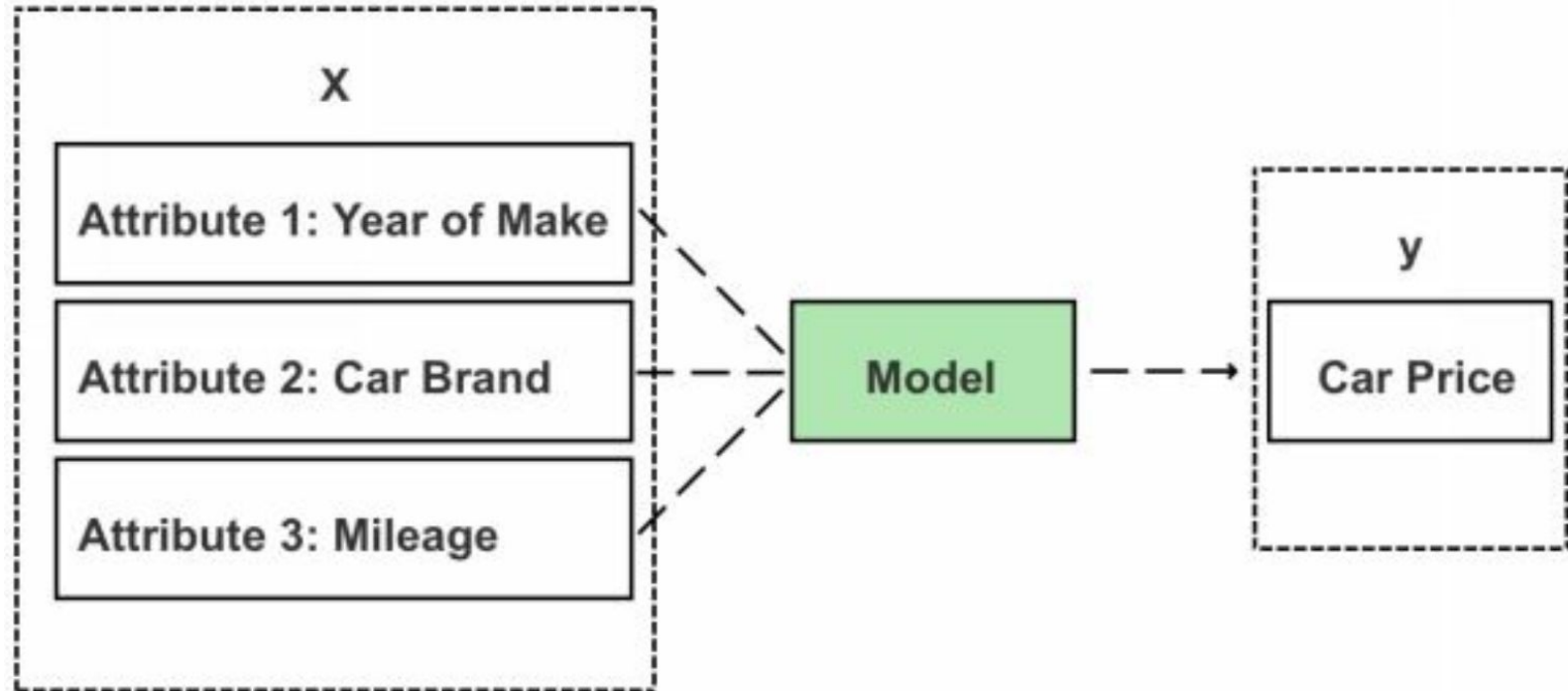
- > Decision process
- > Reward system
- > Learn series of actions

Supervised Learning

- The training set you feed to the algorithm includes the desired solutions, called **labels**.



Supervised Learning



Supervised Learning

	Vector	Matrices		
	Maker (X)	Year (X)	Model (X)	Price (y)
Row 1				
Row 2				
Row 3				
Row 4				



Supervised Learning

- In Machine Learning an **attribute** is a data type
 - (e.g., “mileage”),
- while a **feature** has several meanings, depending on the context, but generally means an attribute plus its value
 - (e.g., “mileage = 15,000”).
- Many people use the words attribute and feature interchangeably.

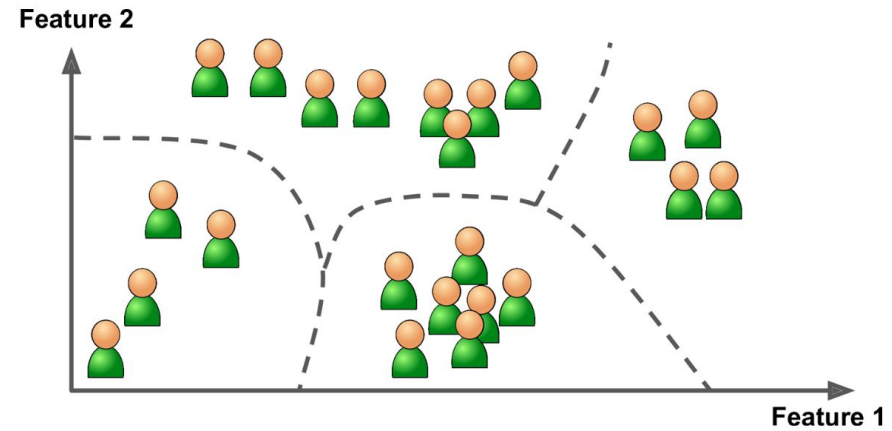
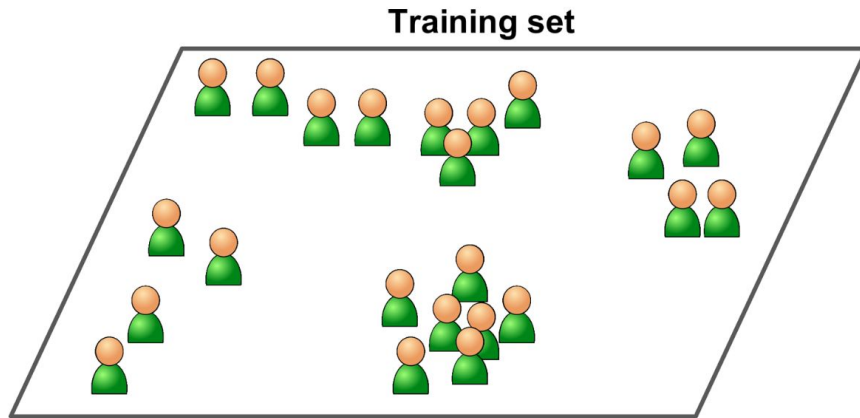


Supervised Learning

- Here are some of the most important supervised learning algorithms.
 - k-Nearest Neighbors
 - Linear Regression
 - Logistic Regression
 - Support Vector Machines (SVMs)
 - Decision Trees and Random Forests
 - Neural Networks

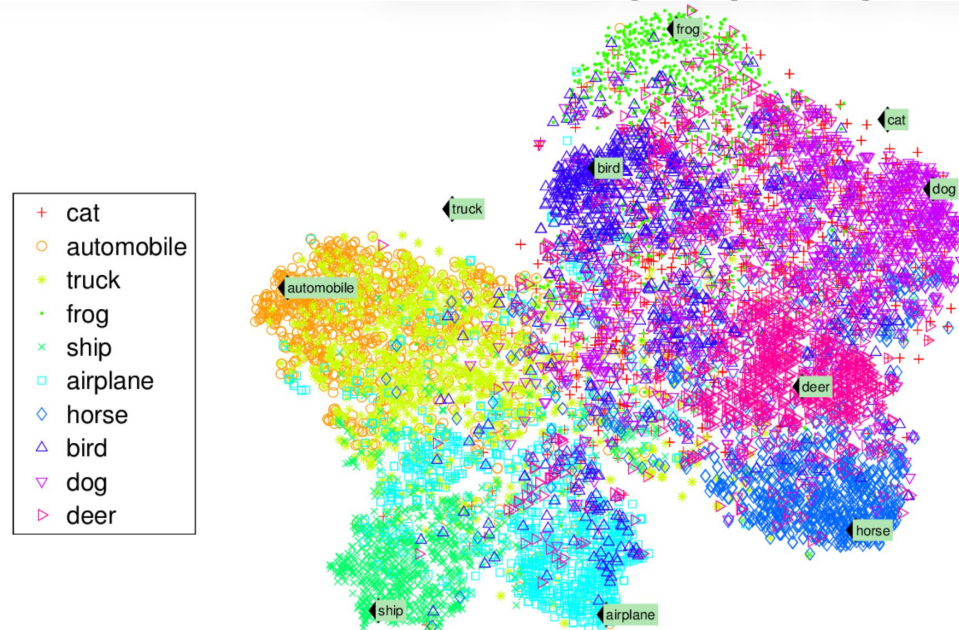
Unsupervised Learning

- The training data is **unlabeled**, the system tries to learn without a teacher.



Unsupervised Learning

- Example of a t-SNE visualization highlighting semantic clusters.





Unsupervised Learning

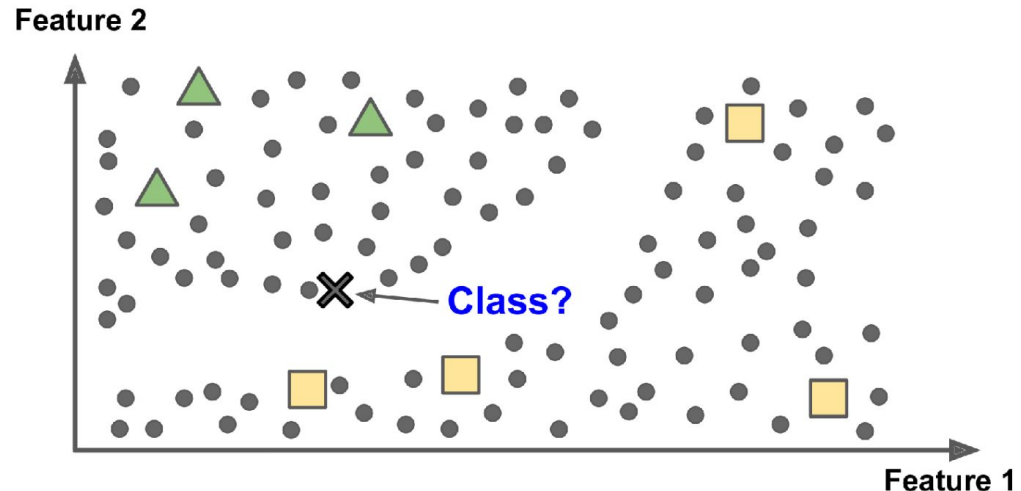
- A related task is **dimensionality reduction**, in which the goal is to simplify the data without losing too much information.
- One way to do this is to merge several correlated features into one, this is called **feature extraction**.
- *It is often a good idea to try to reduce the dimension of your training data using a dimensionality reduction algorithm before you feed it to another Machine Learning algorithm.*

Unsupervised Learning

- Here are some of the most important unsupervised learning algorithms.
 - Clustering (K-Means, DBSCAN, Hierarchical Cluster Analysis)
 - Anomaly detection and novelty detection
 - Visualization and dimensionality reduction (PCA, t-SNE)
 - Association rule learning

Semi-supervised Learning

- Since labeling data is usually time-consuming and costly.
- you will often have plenty of **unlabeled instances**, and **few labeled instances**.

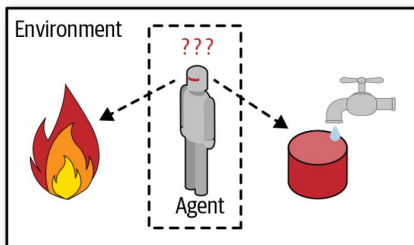




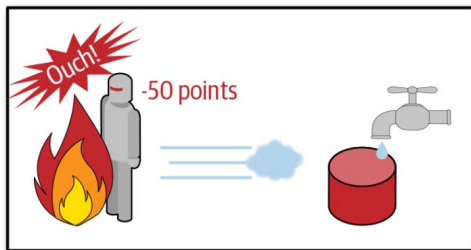
Reinforcement Learning

- The learning system, called an **agent** in this context, can observe the environment, **select and perform actions**, and **get rewards** in return.
- It must then **learn by itself** what is the best strategy, called a **policy**, to get the most reward over time.
- A policy defines what action the agent should choose when it is in a given situation.

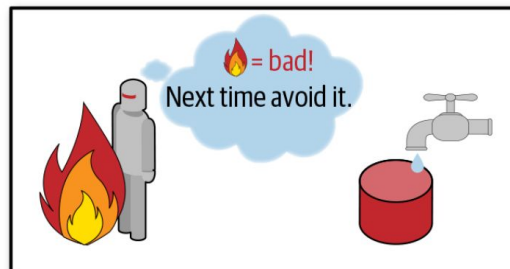
Reinforcement Learning



- 1 Observe
- 2 Select action using policy



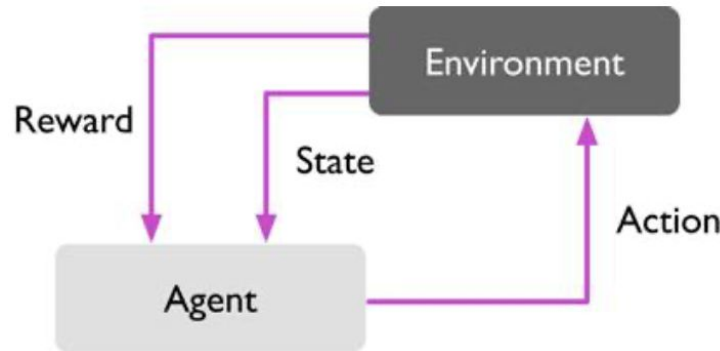
- 3 Action!
- 4 Get reward or penalty



- 5 Update policy (learning step)
- 6 Iterate until an optimal policy is found

Reinforcement Learning

- Through its interaction with the environment, an agent can then use reinforcement learning to learn a series of actions that maximizes this reward via an exploratory **trial-and-error** approach or deliberative planning.



Batch Learning

- In batch learning, the system is **incapable of learning incrementally**.
- It must be trained using all the available data.
- This will generally take a lot of time and computing resources, so it is typically done offline, this is called **offline learning**.

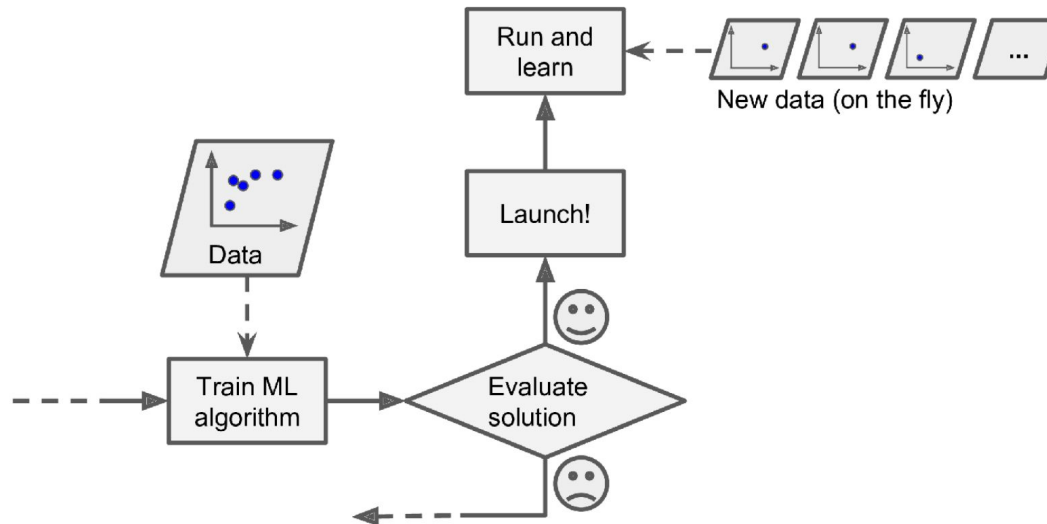


Online Learning

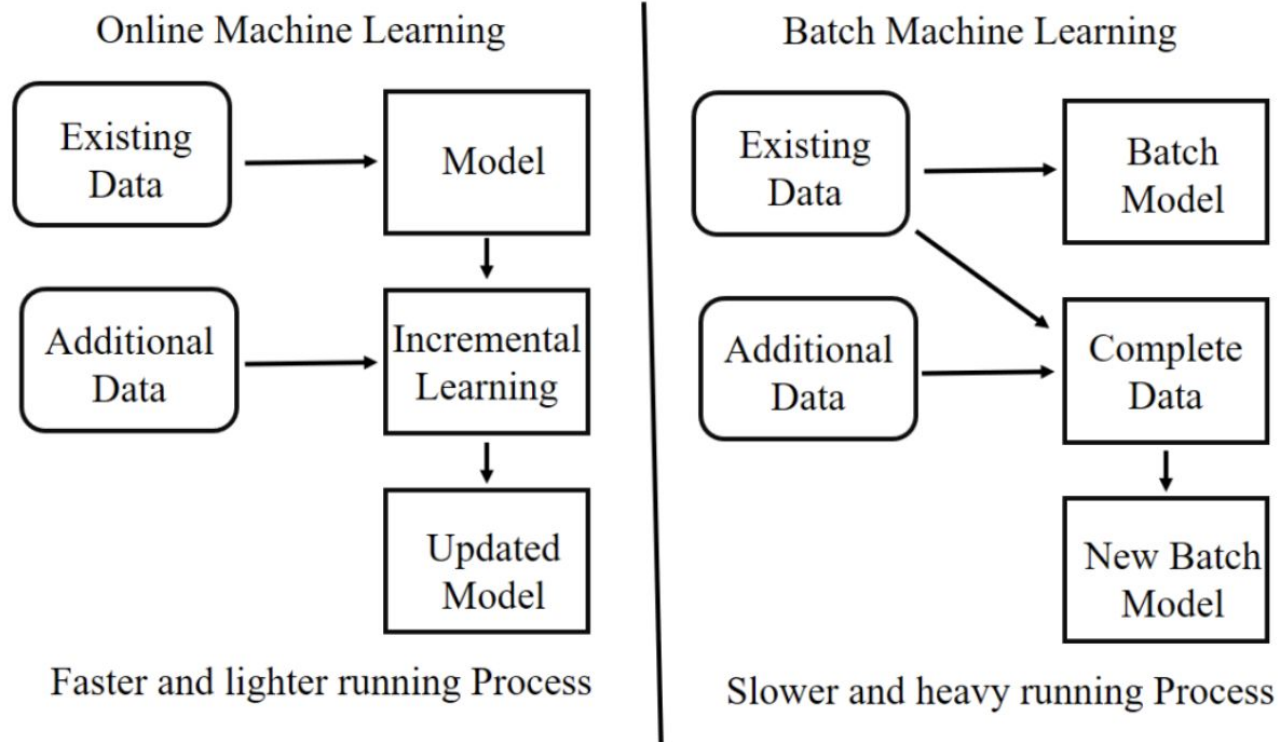
- In online learning, you **train the system incrementally** by feeding it data instances sequentially, either individually or in small groups called **mini-batches**.
- Each learning step is **fast and cheap**, so the system can learn about new data on the fly.

Online Learning

- In online learning, a model is trained and launched into production, and then it keeps learning as new data comes in.



Online Learning





Main Challenges of ML

- The two things that can go wrong are “**bad algorithm**” and “**bad data**.”
 - Insufficient Quantity of Training Data
 - Nonrepresentative Training Data
 - Poor-Quality Data
 - Irrelevant Features (feature engineering)
 - Data Mismatch



Main Challenges of ML

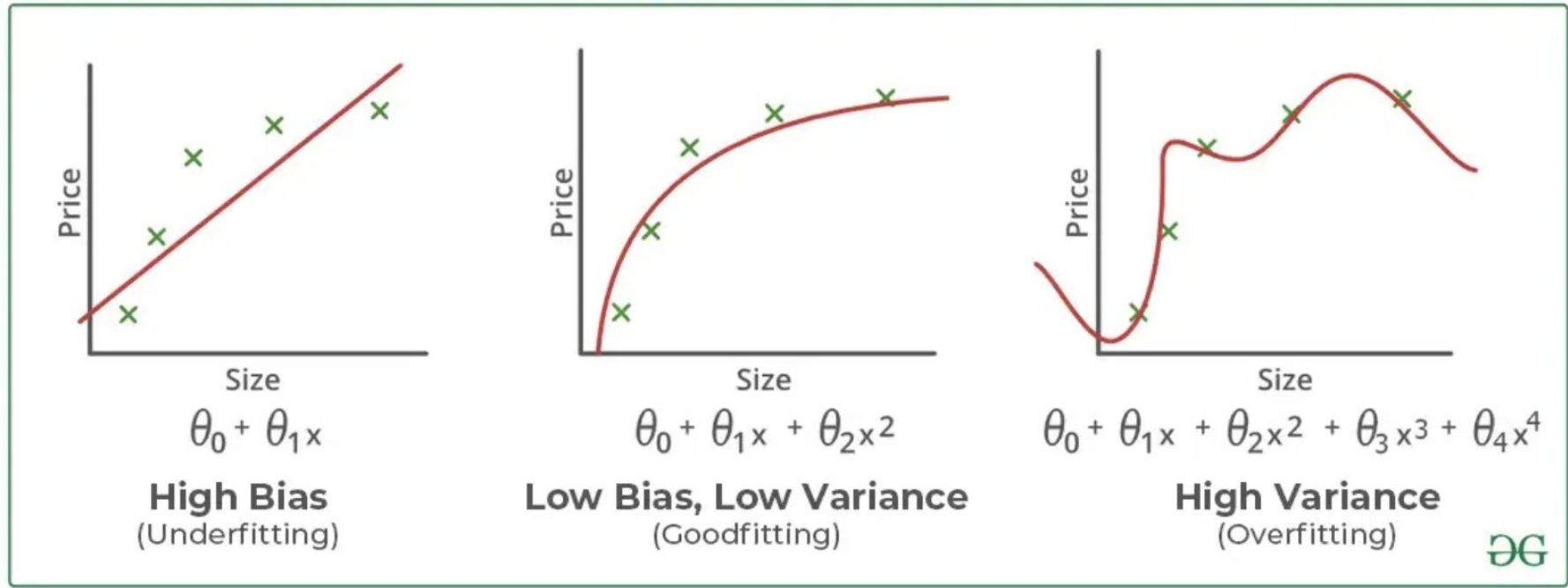
- Overfitting the Training Data
 - It means that the model performs well on the training data, but it does not generalize well.
- Underfitting the Training Data
 - It occurs when your model is too simple to learn the underlying structure of the data, so its predictions are bound to be inaccurate.



Main Challenges of ML

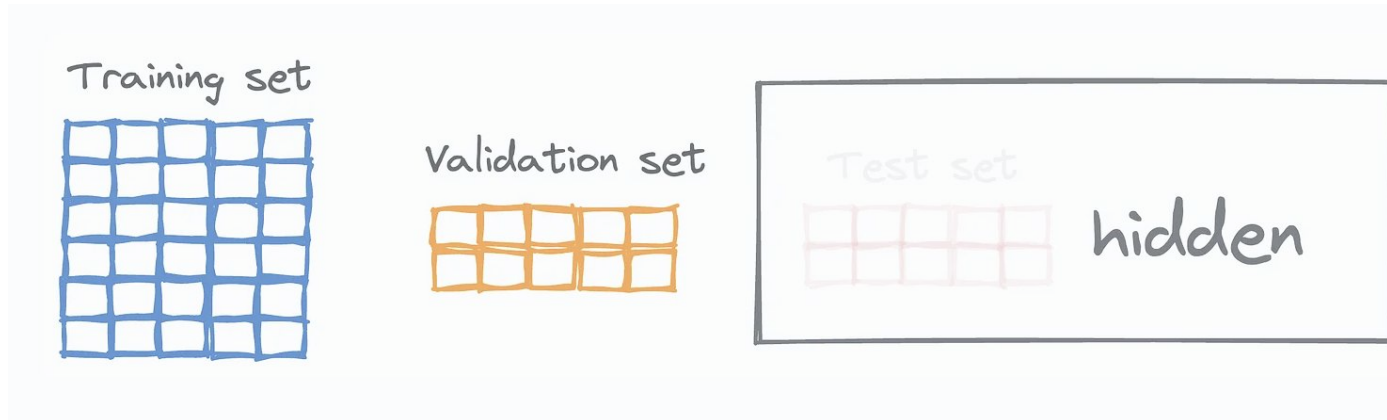
- To understand underfitting and overfitting, you must first understand **bias** and **variance**.
 - **Bias** is the error that happens when a machine learning model is too simple and doesn't learn enough details from the data.
 - **Variance**: Error that happens when a machine learning model learns too much from the data, including random noise.

Main Challenges of ML

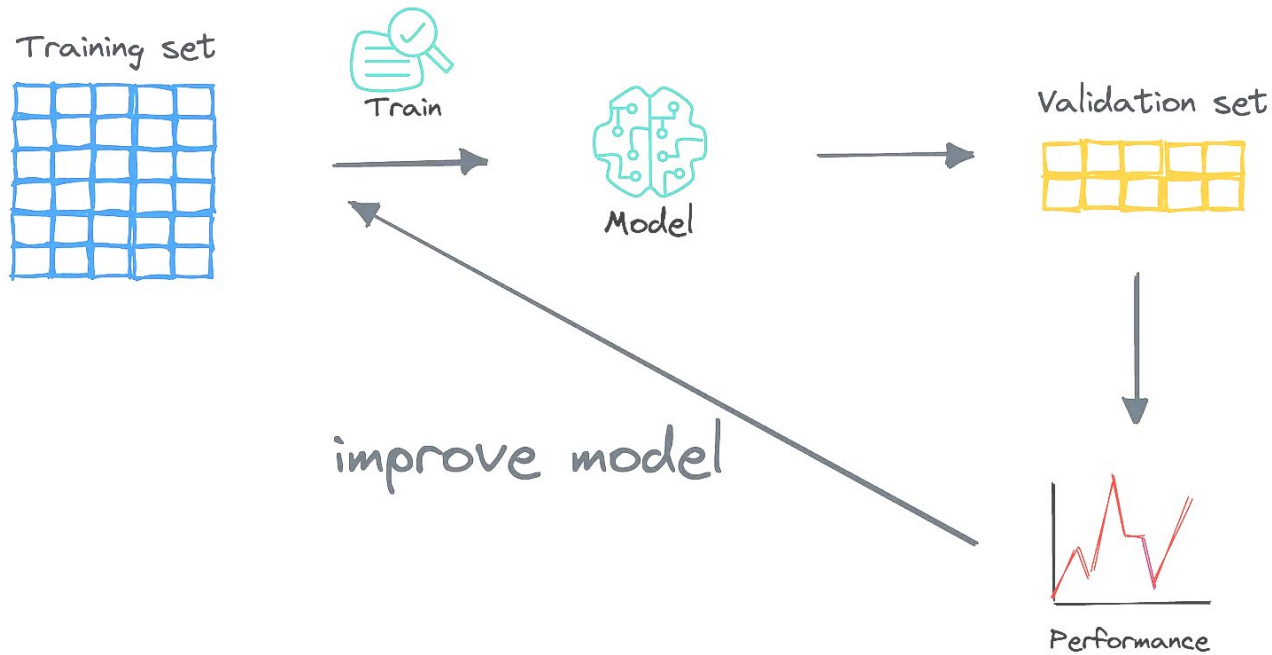


Testing, Validating and Test

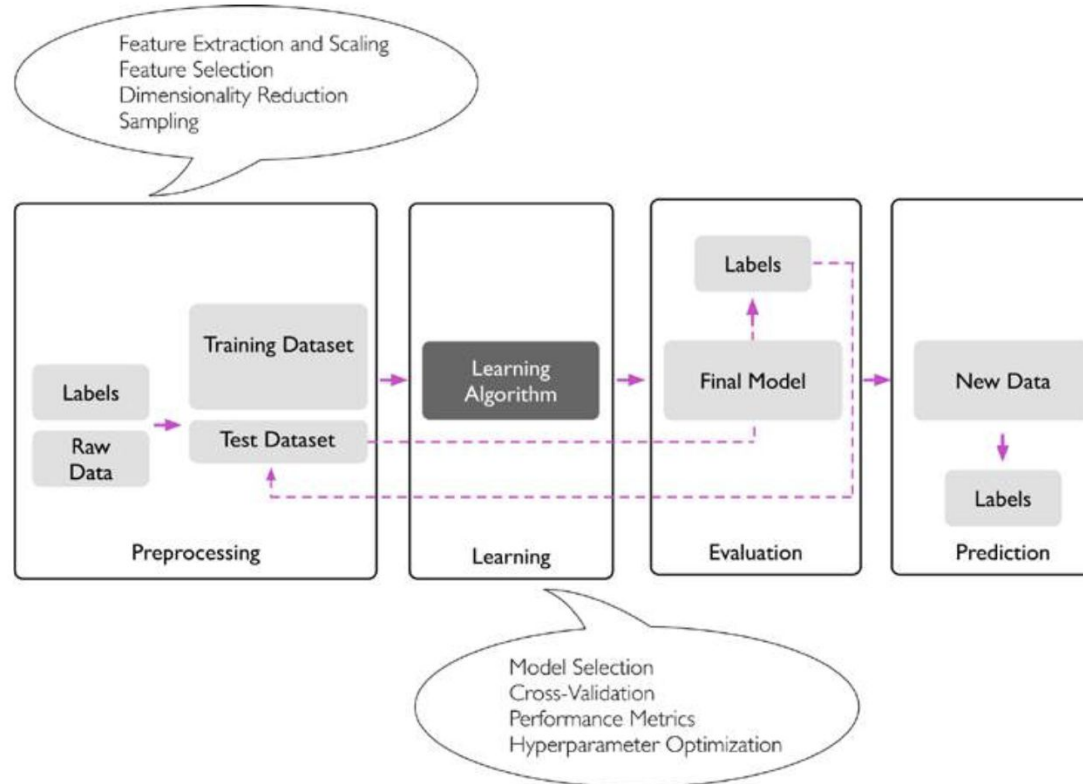
- Split data into two sets: the **training set** and the **test set**.
- Train your model using the training set, and you test it using the test set.



Testing, Validating and Test



A Roadmap for Building ML





A Roadmap for Building ML

- In summary:
 - You studied the data.
 - You selected a model.
 - You trained it on the training data.
 - Finally, you applied the model to make predictions on new cases (this is called inference), hoping that this model will generalize well.



Examples of Applications

- Analyzing images of products on a production line to automatically classify them.
- Detecting tumors in brain scans.
- Automatically classifying news articles.
- Automatically flagging offensive comments on discussion forums.
- Creating a chatbot or a personal assistant.



Examples of Applications

- Detecting credit card fraud.
- Building an intelligent bot for a game.
- Making your app react to voice commands.
- Segmenting clients based on their purchases so that you can design a different marketing strategy for each segment.



End

- Q&A
- Thank you :)